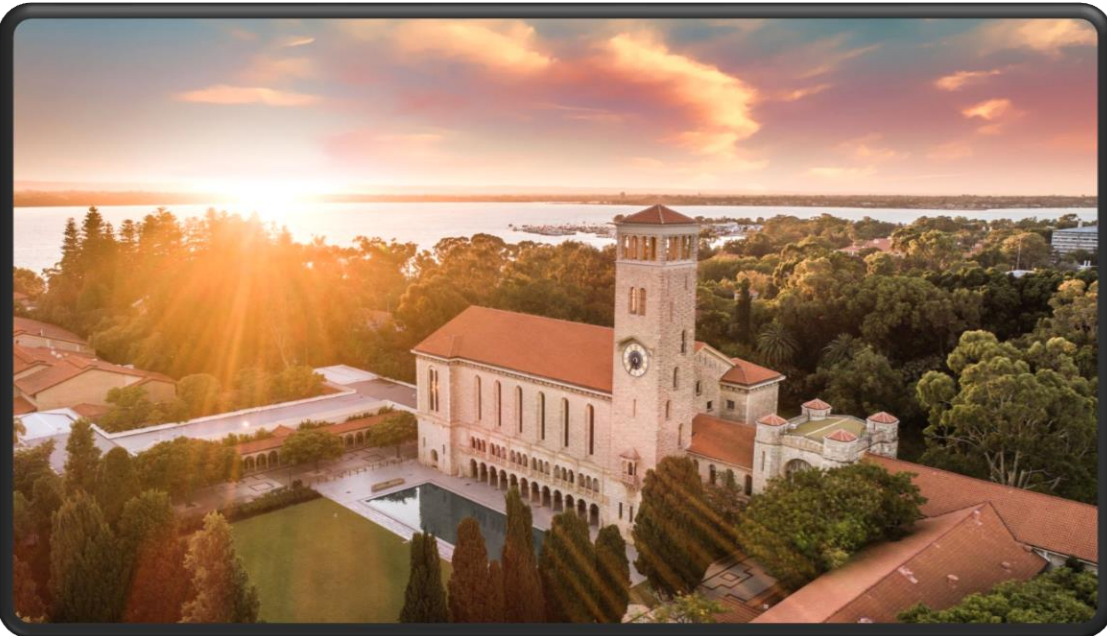


INMT5518

Supply Chain Analytics

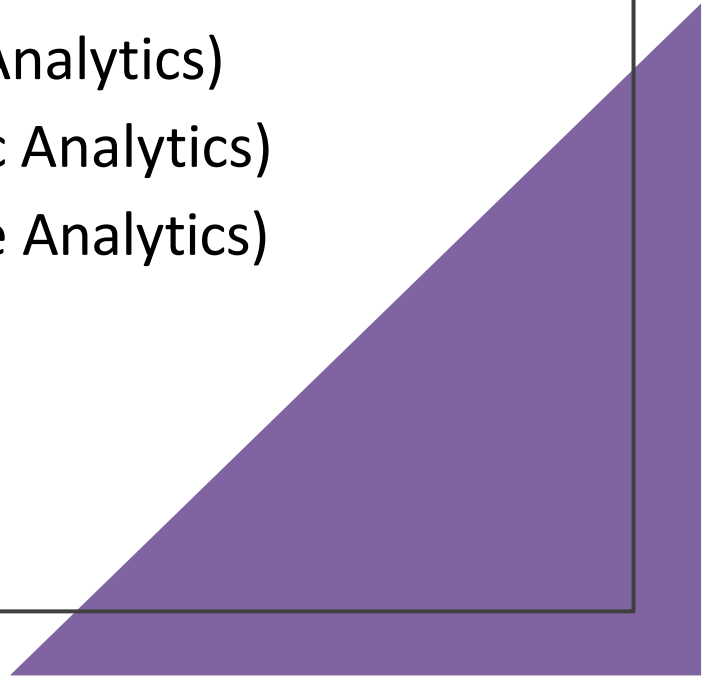
Week 3: Basic Tools

Presented by:
Dr. Mehdi Rajabi Asadabadi

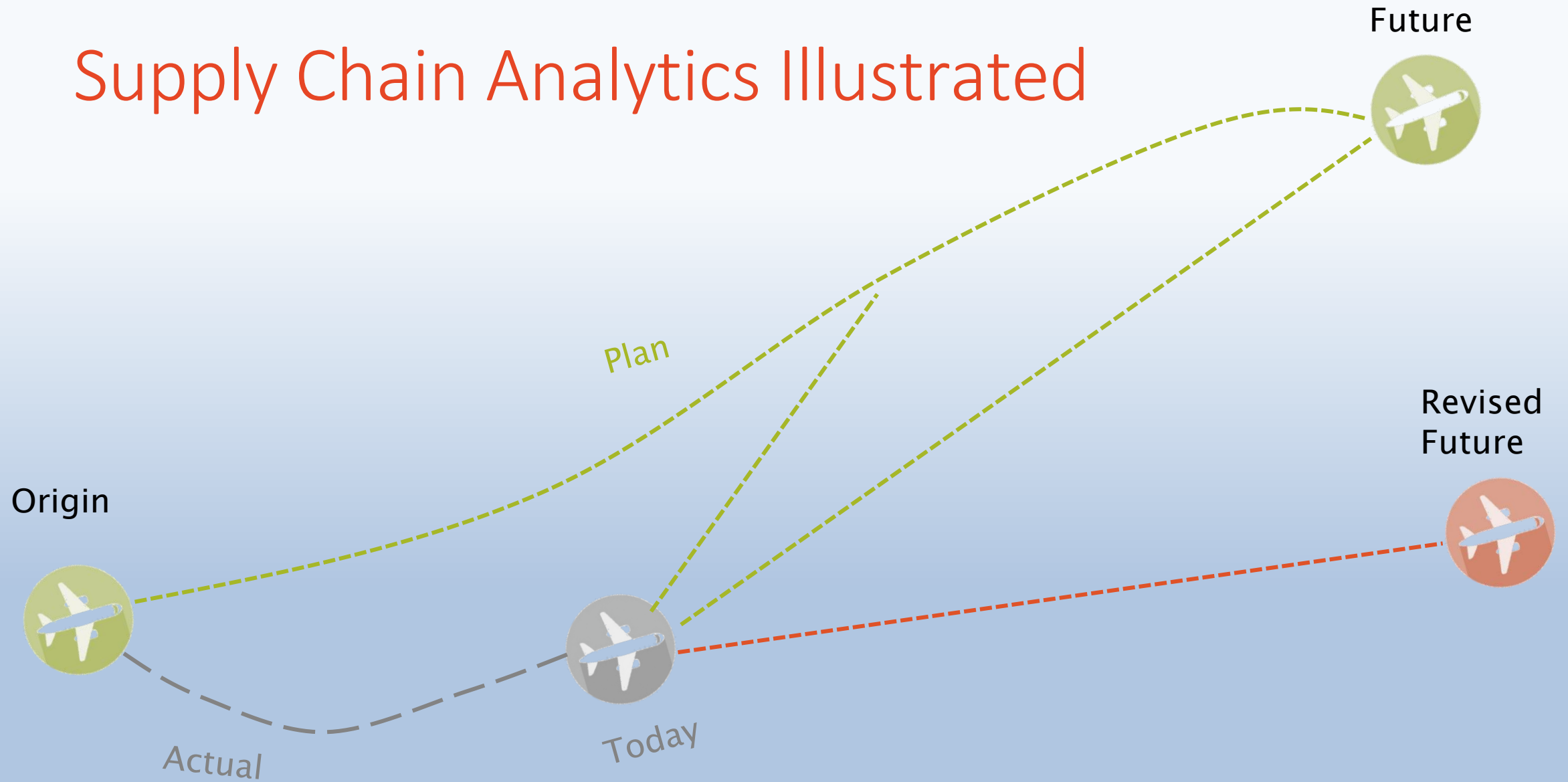


Five Key Questions in Supply Chain Analytics

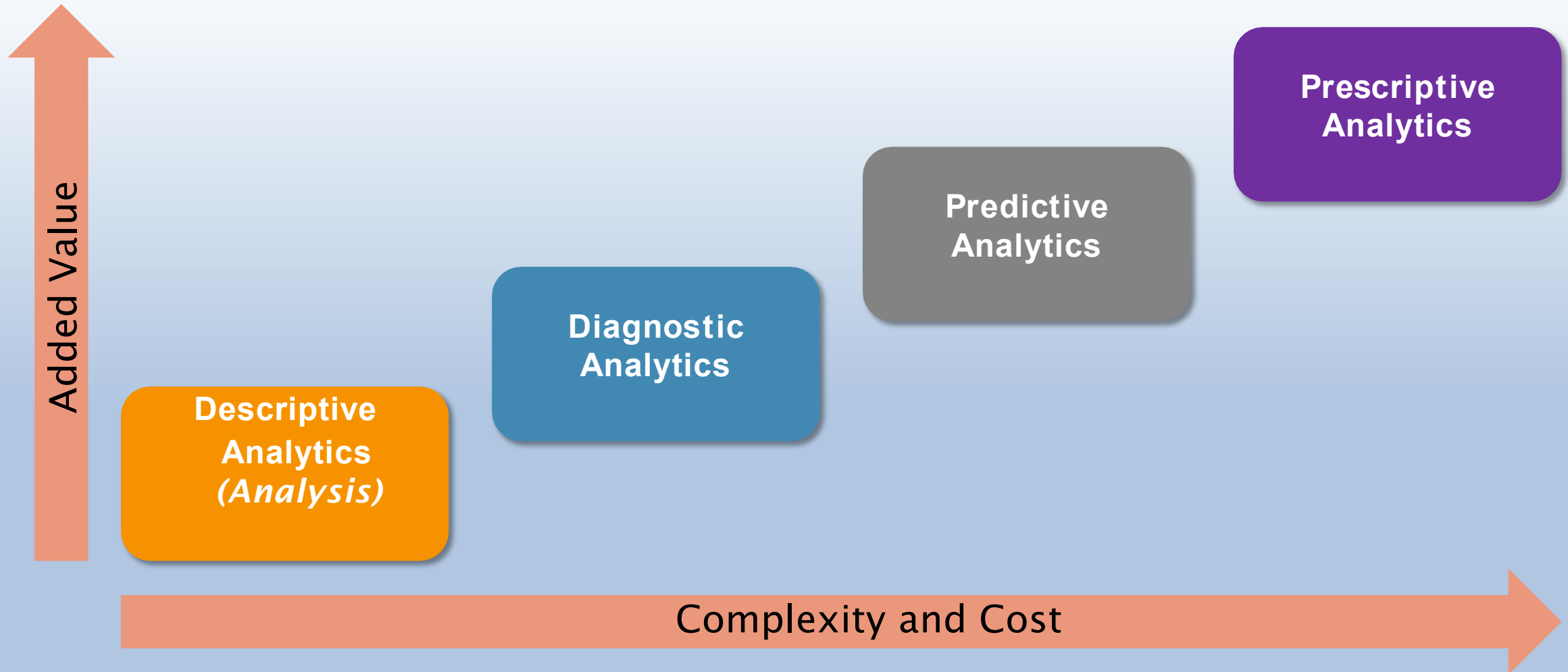
- What is my plan? (The pathway to achieve your goal)
- What is my present position (where am I)? (Descriptive Analytics)
- What are the variances? What caused them? (Diagnostic Analytics)
- What are the trends? What are the forecasts? (Predictive Analytics)
- What actions are required? (Prescriptive Analytics)



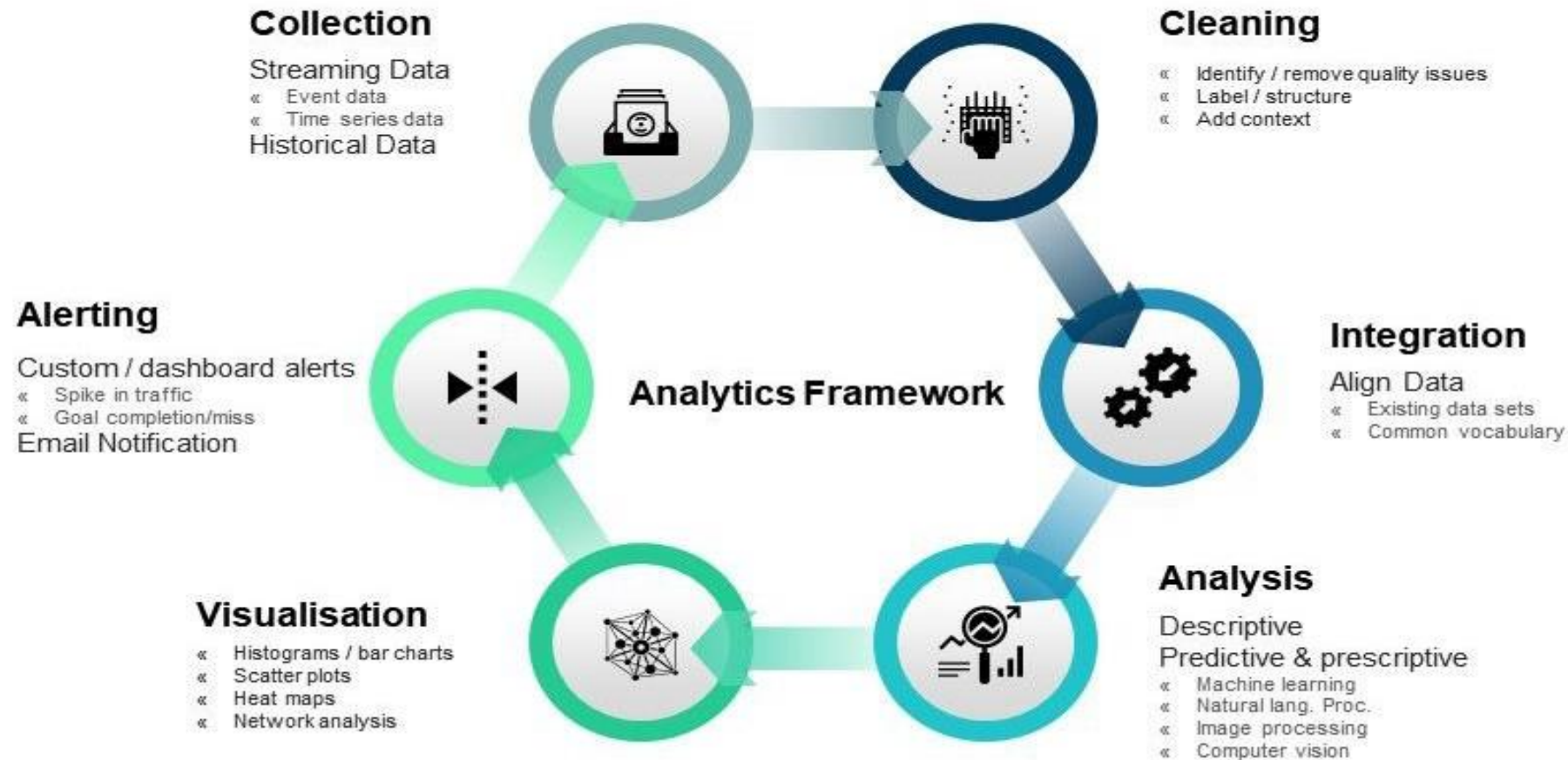
Supply Chain Analytics Illustrated



Types of Analytics



Data analytics cycle



Structured and Unstructured Data

V22																
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
1	Customer_mou_Meatotmrc_M	rev_Range	mou_Ran	change_m	drop_blk	drop_vce	owylis_vc	mou_opk	months	totcalls	eqpdays	custcare	lcallwait	nwiylis_vc		
2	1064525	190.25	63.94	26	43	-11.25	4.666667	8	20	28.44	14	1104	403	0	0.666667	10.33333
3	1048538	443	39.99	5.1	199	-78	4.333333	5	33	72.46	13	2237	404	1.333333	0	16.33333
4	1010139	400.5	44.99	13.88	172	-67.5	2	1	7	93.6	29	3276	213	1	0	0
5	1014496	53.5	34.6675	18.56	78	12.5	1.333333	2	0	4.7	25	1932	757	4.666667	0	0
6	1012053	37	21.0425	35.79	74	-33	5.666667	1	4	36.6	33	652	983	0.333333	0	0
7	1028421	1181.5	44.99	62.3	381	-67.5	5.666667	2	40	332.92	25	5122	273	1	0	30.66667
8	1061694	543.5	69.99	16.94	462	326.5	13	3	8	5.48	16	3138	391	0	1.333333	1.333333
9	1038047	1463.25	84.99	97.03	1280	-824.25	28.33333	19	17	466.76	19	5456	283	12	3.333333	11.33333
10	1084670	93.5	65.475	15.96	23	9.5	1	1	2	22.5	9	314	258	2	0	0.333333
11	1096350	1225.5	35.99	4.2	107	10.5	10	5	11	7.43	8	3441	215	1.666667	11.66667	2
12	1034639	1196.5	39.99	63.65	82	57.5	10.33333	2	18	91.29	22	4289	655	1	4.666667	7.333333
13	1009776	610	49.99	131.46	694	-315	13.66667	11	9	140.17	35	11307	228	0	0	1.333333
14	1059674	1354	84.99	37.51	506	-173	12	2	90	137.72	15	7709	430	1.333333	10	38
15	1012178	787.5	44.615	303.01	975	107.5	12	6	47	104.29	31	7870	311	1.333333	6.666667	6
16	1051763	233	39.99	35.71	309	-130	11.33333	13	7	29.88	12	1151	116	0.333333	0.666667	0
17	1028780	263.5	30	0.54	29	-16.5	5.333333	6	6	46.94	19	1636	578	0	0	1.666667
18	1082824	1054.75	44.99	33.67	588	186.25	17.33333	5	50	246.08	7	2356	201	1.333333	1	10.66667
19	1005955	652	49.99	39.77	156	49	7.666667	4	18	68.02	38	10420	351	1.333333	0.666667	1
20	1062893	12	10	3.6	36	24	0	0	0	0	12	151	366	0	0	0

2.1 Introduction

A metadata service for business interactions provides information about what kinds of data transactions, and what kinds of enabling technologies for those transactions, are available for specific business process participants. While a web site that offers natural language "implementation guides" counts as a metadata service broadly viewed, in this document the focus will be on approaches that have emerged to automate the setup and life-cycle management of business interactions.

The location of a metadata service in this document refers primarily to a URL-specified endpoint identifier [RFC3986,2616]. For the location of a human-readable and web-browsable document, a link in an email invitation, on a web page or in the results of a web search engine might provide the service location information. For an automated process, however, it is desirable to have a specific way to publish and retrieve location information. Ideally, the procedure for metadata service location would be as reliable and pervasive as the ability to find the numerical IP address for a given path of domain name system labels. Interestingly enough, the DNS system itself has specified record types that can store URLs [RFC2915,RFC4848], and whose use in building a location finding infrastructure for service location (of any kind) has been specified [RFC3401,3402,3403,3404].

The goal of this specification is to provide ways to find the location of a metadata service of a given type, to enable the use of information provided by that service to setup and start-up business interactions. Some specific services will be provided with standard designated values for use in DNS records; patterns for values of the DNS service field are provided to enable extension points for future community implementation standardizations or other standards bodies defining metadata.

2.2 Overview of the Core Service Location Discovery System

The goal of a Dynamic Delegation Discovery System (DDDS) application for metadata service discovery is to find URLs of specific types of metadata services, by using a DNS query string that represents an identity of a person or organization. In other words, the goal is to retrieve one or more URLs from the DNS that will enable finding out more about an entity's enabled metadata services.

The framework for such a DDDS application sets out what kinds of information must be specified:

- A DNS query string which is a string of concatenated, dot-separated labels.
- Types of DNS records to be retrieved.
- Values for fields within DNS record types that are used.
- Processing steps to be taken.

Because the goal is to find URLs, the NAPTR RR with "U" value in its Flags field (U-NAPTR), provides the core framework for producing the desired DDDS application [RFC4848]. The service field of a U-NAPTR can distinguish distinct kinds of metadata services, or supporting services such as registration services to obtain access authorization for metadata services. DNS domain names of organizations or persons provide the core solution for query strings that may be used when trying to locate metadata services for persons or organizations.

So when "example.com" wants to announce the URL for metadata services, its IT services group can add a U-NAPTR record configuration such as:

```
IN NAPTR 100 10 "U" "Meta:CPA" "!" ".*$https://example.com/cpa!" .
```



Ambiguous requirements: A semi-automated approach to identify and clarify ambiguity in large-scale projects

Mehdi Rajabi Asadabadi ^{a, b, 1} , Morteza Saberi ^{c, 2} , Ofer Zwikael ^{a, 3} , Elizabeth Chang ^{b, 4}

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<https://doi.org/10.1016/j.cie.2020.106828>

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Highlights

- Ambiguity in setting product requirements is a major problem in large scale projects.
- This problem often leads to poor project selection decisions

The current issue and full text archive of this journal is available on Emerald Insight at:
<https://www.emerald.com/insight/1741-0398.htm>

Enhancing the analysis of online product reviews to support product improvement: integrating text mining with quality function deployment

Reviews to
support
product
improvement

275

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*School of Information and Communication Technology, Griffith University,
Queensland, Australia*

Abstract

Purpose – The purpose of this paper is to develop an effective approach to support and guide production improvement processes utilising online product reviews.

Design/methodology/approach – This paper combines two methods: (1) natural language processing (NLP) to support advanced text mining to increase the accuracy of information extracted from product reviews and (2) quality function deployment (QFD) to utilise the extracted information to guide the product improvement process.

Findings – The paper proposes an approach to automate the process of obtaining voice of the customer (VOC) by performing text mining on available online product reviews while considering key factors such as the time of review and review usefulness. The paper enhances quality management processes in organisations and advances the literature on customer-oriented product improvement processes.

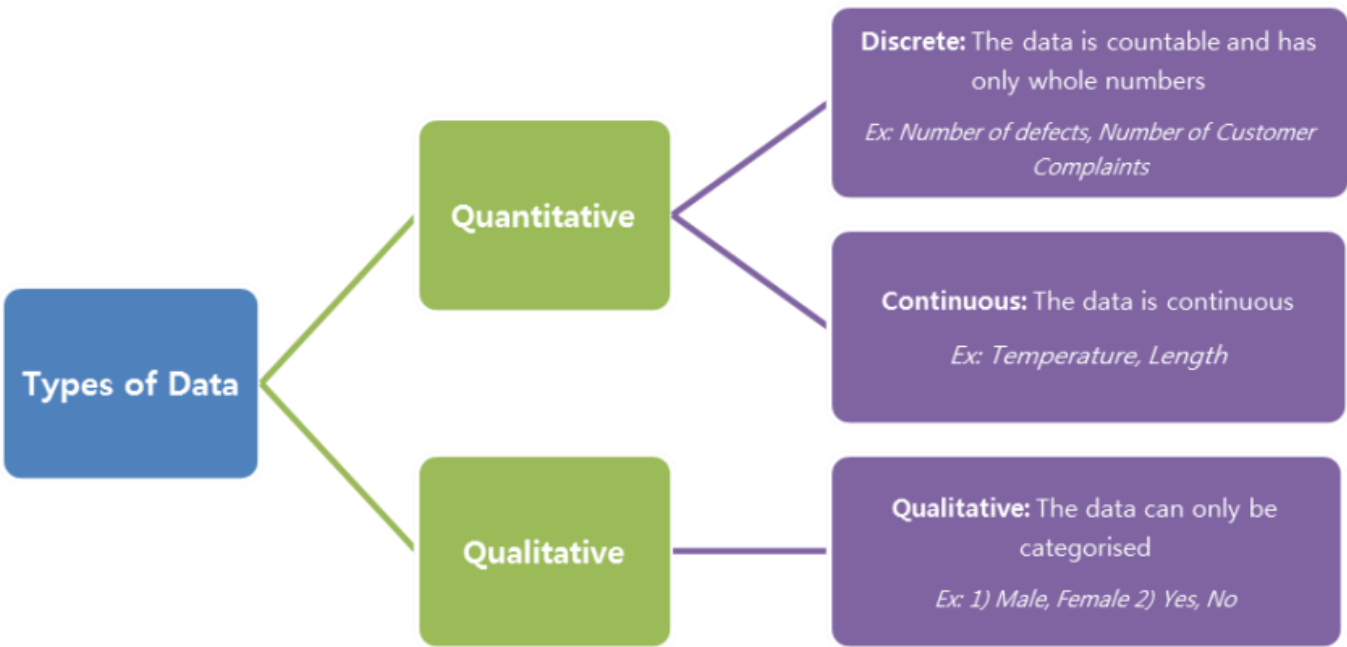
Originality/value – Online product reviews are a valuable source of information for companies to capture the true VOC. VOC is then commonly used by companies as the main input for QFD to enhance quality management and product improvement. However, this process requires considerable time, during which VOC may change, which may negatively impact the output of QFD. This paper addresses this challenge by providing an improved approach.

Keywords Customer requirements, Online product reviews, QFD, Voice of the customer

Paper type Research paper

Overarching theme	Sub-theme and coding instructions	Sample of coded text
Initiation into playing bingo	The first time they heard of bingo or knew of someone who played bingo.	Remembers when she was 14 going down to meet her mother at Gala bingo.
	Family and friends history of playing bingo	Mother went to bingo, had uncles that gambled on horses, father wasn't a gambler, only gambled on a big race like Grand National
	Their first bingo experience	First time she played bingo was at the Old Vogue with her mum aged 18.
On-line bingo appeal	Frequency and duration of play	She hasn't been to bingo for 6 months, but now plays online bingo. Has done so for 4 months.
	Reasons for playing/likes and dislikes of playing	Likes online bingo because if you play with £10 they give you an additional £5 to play with.
	Strategy for playing	She has her shower, moisturises, puts a face pack on, fresh pyjamas, has a glass of wine, a smoke & plays bingo online. She sits & chats with the others.
Variety in bingo games and other types of gambling.	Type of gambling	She'd heard about the online bingo from a couple of women at the bingo. She recently tried a different online bingo run by the Sunday Mail. She got £10 free without having to put any money in herself. She played with that £10 & has not been on since as didn't like it. Found it tacky & less friendly than the STV one she is used to. (13)

doi:10.1371/journal.pone.0154763.t003



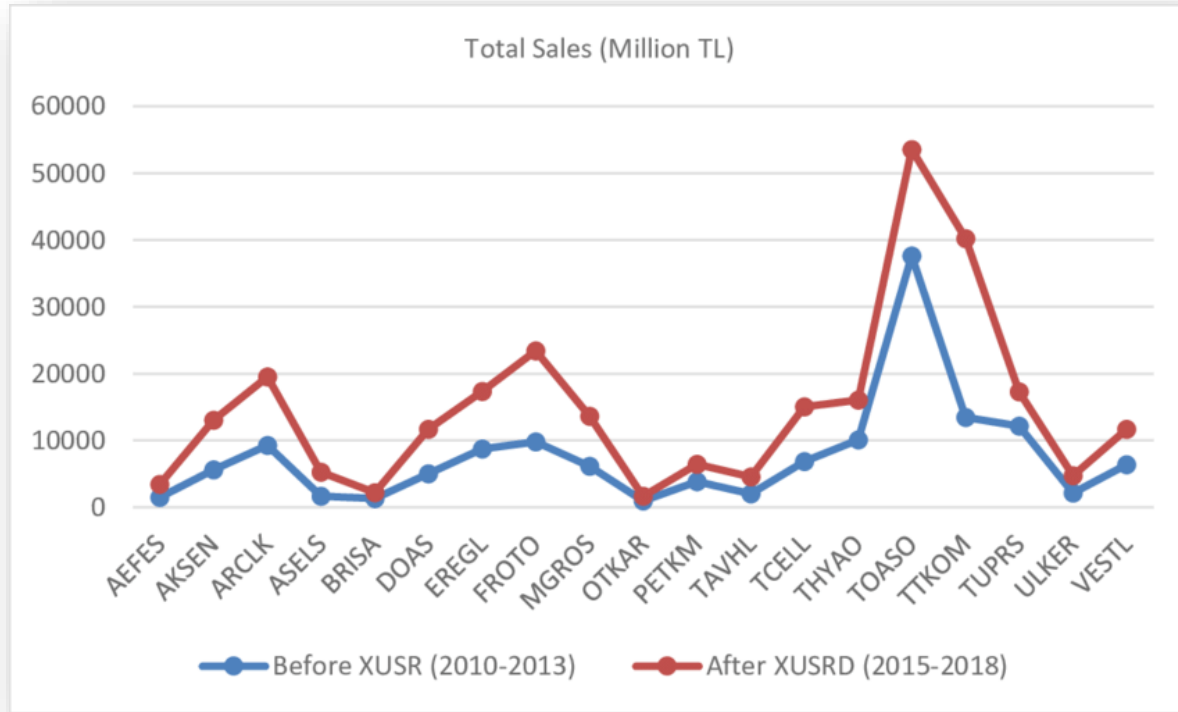
V22																		
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P			
Customer	mou	Mea	totmrc	M_rev	Range	mou_Ran	change_m	drop_blk	drop_vce	owylis_vc	mou_opk	months	totcalls	eqpdays	custcare	callwait	Ni	owylis_vce
2	1064525	190.25	63.94	26	43	-11.25	4.666667	8	20	28.44	14	1104	403	0	0.666667	10.33333		
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7	1028421	1181.5	44.99	62.3	381	-67.5	5.666667	2	40	332.92	25	5122	273	1	0	30.66667		
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20	1062893	12	10	3.6	36	24	0	0	0	0	12	151	366	0	0	0		

Which type of data is mostly used in Supply Chain Analytics, Qualitative or Quantitative?

Overarching theme	Sub-theme and coding instructions	Sample of coded text
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Format Painter Clipboard Font Alignment Number Formatting Table																
V22																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
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Tables and Figures



☐		* Task	Owner	* Priority	* Status	Deadline
☐		Organize conference agenda	Eric Wallace	High	Finished	2021
☐		Prepare conference content	Lucas FADEL	Medium	Finished	2021
☐		Reserve rooms and catering	Lucas FADEL	High	In progress	2021
☐		Send Save the Date email	Robert Cantu	High	Finished	2021
☐		Print conference materials	Robert Cantu	Low	In progress	2018
☐		Send official invitation	Eric Wallace	High	To do	2021
☐		Prepare report for team me...	Leila KUPHAL	Medium	Finished	2021
☐		Organize team event	Eric Wallace	Low	In progress	2021

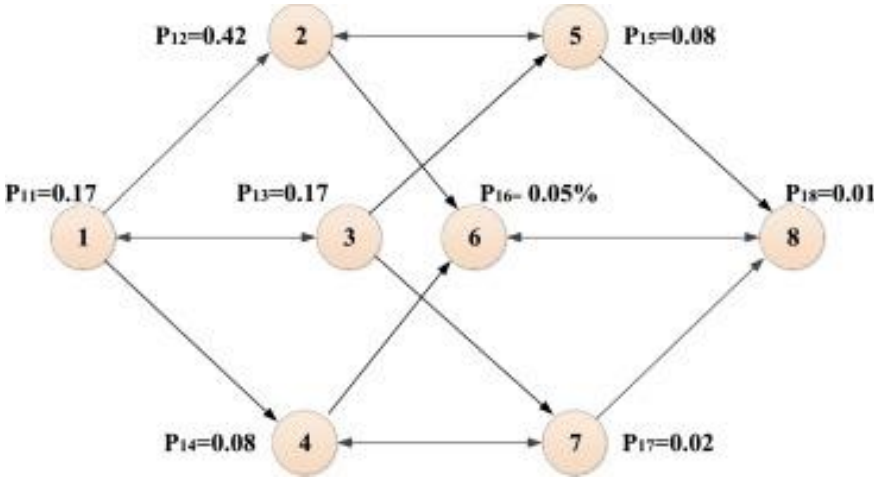
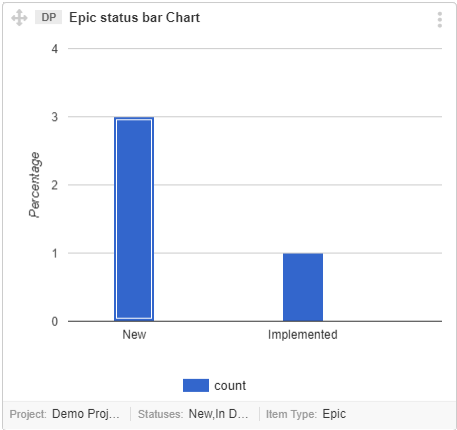
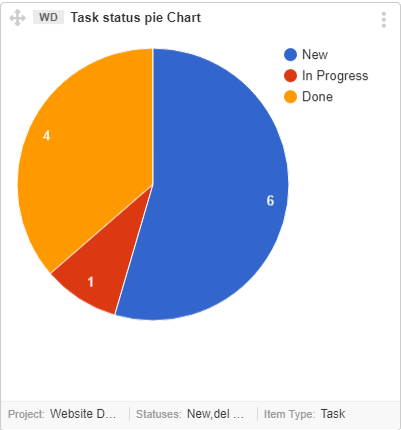
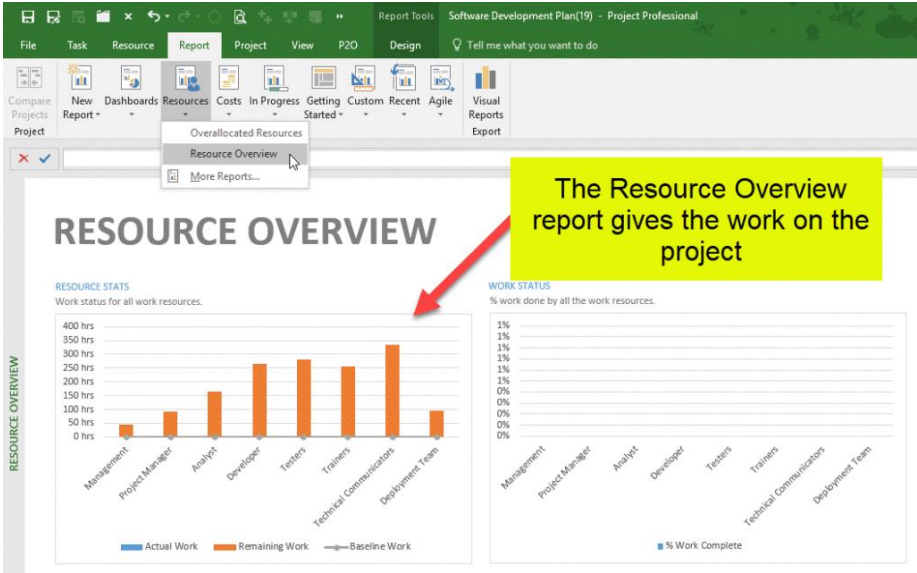
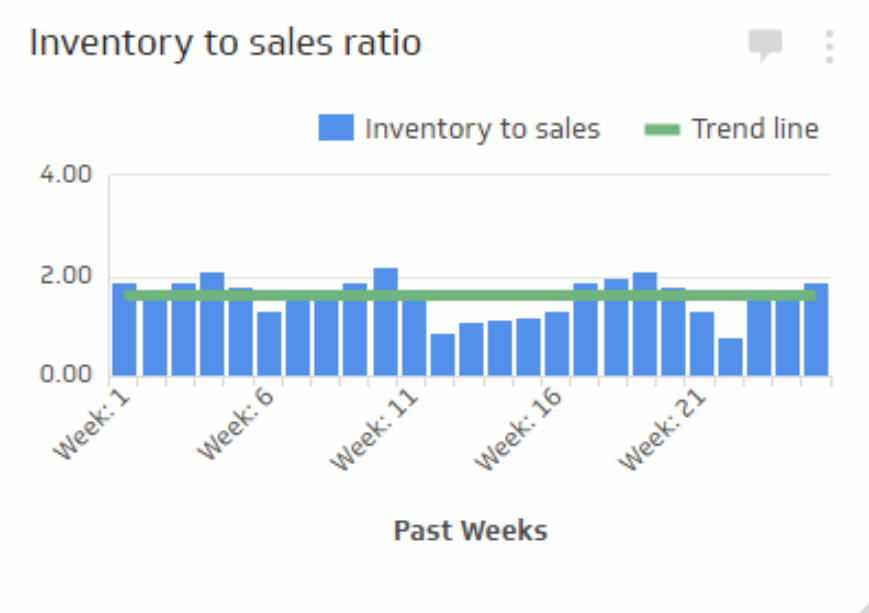
Table

☐		* Task	  Owner	* Priority	* Status	 Deadline
☐	EW	Organize conference agenda	 Eric Wallace	High	Finished	 202
☐	LF	Prepare conference content	 Lucas FADEL	Medium	Finished	 202
☐	LF	Reserve rooms and catering	 Lucas FADEL	High	In progress	 202
☐	RC	Send Save the Date email	 Robert Cantu	High	Finished	 202
☐	RC	Print conference materials	 Robert Cantu	Low	In progress	 201
☐	EW	Send official invitation	 Eric Wallace	High	To do	 202
☐	LK	Prepare report for team me...	 Leila KUPHAL	Medium	Finished	 202
☐	EW	Organize team event	 Eric Wallace	Low	In progress	 202

Matrix

		Impact →				
		Negligible	Minor	Moderate	Significant	Severe
Likelihood ↑	Very Likely	Low Med	Medium	Med Hi	High	High
	Likely	Low	Low Med	Medium	Med Hi	High
	Possible	Low	Low Med	Medium	Med Hi	Med Hi
	Unlikely	Low	Low Med	Low Med	Medium	Med Hi
	Very Unlikely	Low	Low	Low Med	Medium	Medium

Graphs and Charts



Popular Analytics Tools

- Microsoft Excel
- Python
- R
- Jupyter Notebook
- Apache Spark
- SAS
- Microsoft Power BI
- Tableau
- KNIME



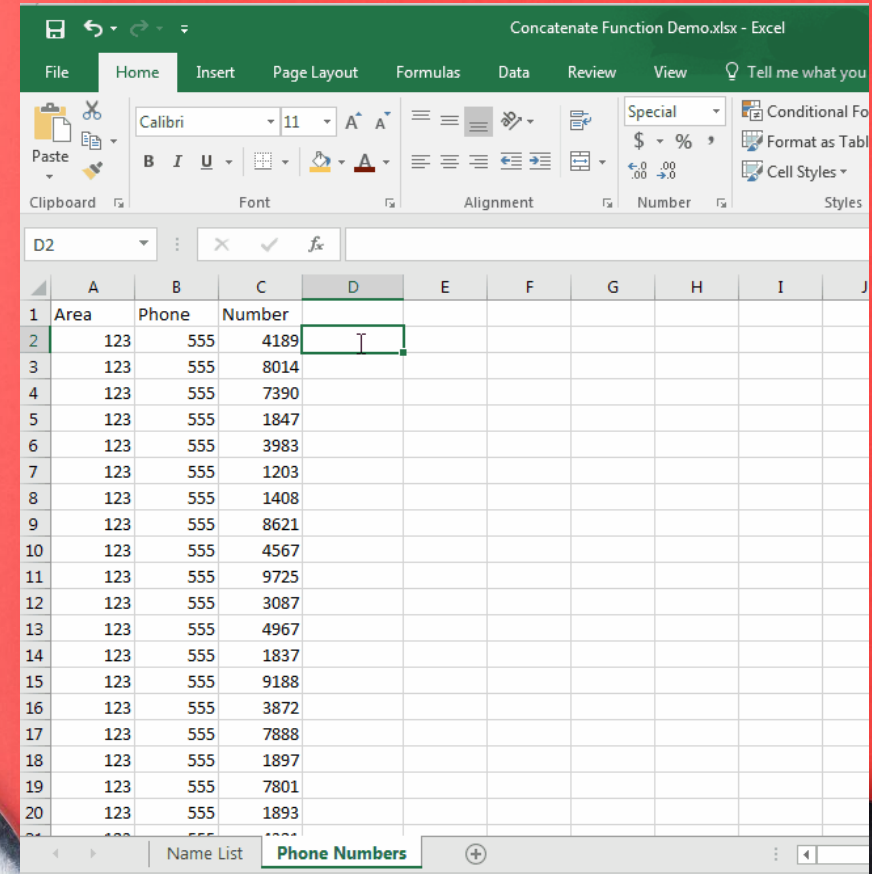
Popular Analytics Tools

Excel is the world's best-known **spreadsheet-based** software.

It is a **popular tool for data cleaning and preparing reports.**

Merits:

Widely-used, lots of useful functions, plug-ins and graphing options



Popular Analytics Tools

Python is a programming language.

It is **open source**, so it is free, with thousands of free libraries.

For example, if you want to do some sort of sentiment analysis, then you find a library that does this, and then provoke or call that library.

It is relatively **easy to learn** and is widely-used.

In general, if you're **building software from scratch**, then Python is probably the best option to use.

script.py

```
1 facebook="Facebook's rating is"
2 fb_rating=3.5
3 fb_rating_str=str(3.5)
4 fb=facebook+' '+fb_rating_str
5 print(fb)
```

```
mine) C:\Users\Mehdi\Desktop\SMCDM-main>python main.py -i templates/input.csv -e templates/events.csv -d out.png
00%|          | 6/6 [00:00:00:00, 2994.86it/s]
00%|          | 6/6 [00:00:00:00, 2994.86it/s]
00%|          | 0/26 [00:00:00:00, 2994.86it/s]
.8999999999999999 < 1
likelihood below the threshold. Skipping the combination for {'B', 'D', 'C', 'A'}
.5249999999999999 < 1
likelihood below the threshold. Skipping the combination for {'B', 'D', 'A', 'E'}
.7875 < 1
likelihood below the threshold. Skipping the combination for {'B', 'D', 'C', 'E'}
88%|          | 23/26 [00:00:00:00, 920.43it/s]
00%|          | 20/20 [00:00:00:00, 920.43it/s]
00%|          | 26/26 [00:00:00:00, 920.43it/s]
00%|          | 26/26 [00:00:00:00, 920.43it/s]
00%|          | 26/26 [00:00:00:00, 920.43it/s]
00%|          | 26/26 [00:00:00:00, 920.43it/s]

final results table:
  option  cr01  cr02  cr03  results
1         5     7     2    388.100665
2         5     3    20    832.770865
3         9     5     2    464.084139
4         2     2     8    352.714350
5         1     5     1    183.937159

the best option is :
```

Popular Analytics Tools

R is also an open-source programming language

It is mostly used for **statistical analysis** and **data mining**

It is slower, less secure, and more complex to learn than Python.

So, unless you are doing some sort of **statistical analysis** where you may use R, Python is a better programming language.

code		output
1	<code>a = 1</code>	
2	<code>while a < 7 :</code>	
3	<code> if(a % 2 == 0):</code>	
4	<code> print(a, "is even")</code>	
5	<code> else:</code>	
6	<code> print(a, "is odd")</code>	
7	<code>a += 1 Wikitechy.com</code>	
variables		
<code>a</code> 1		

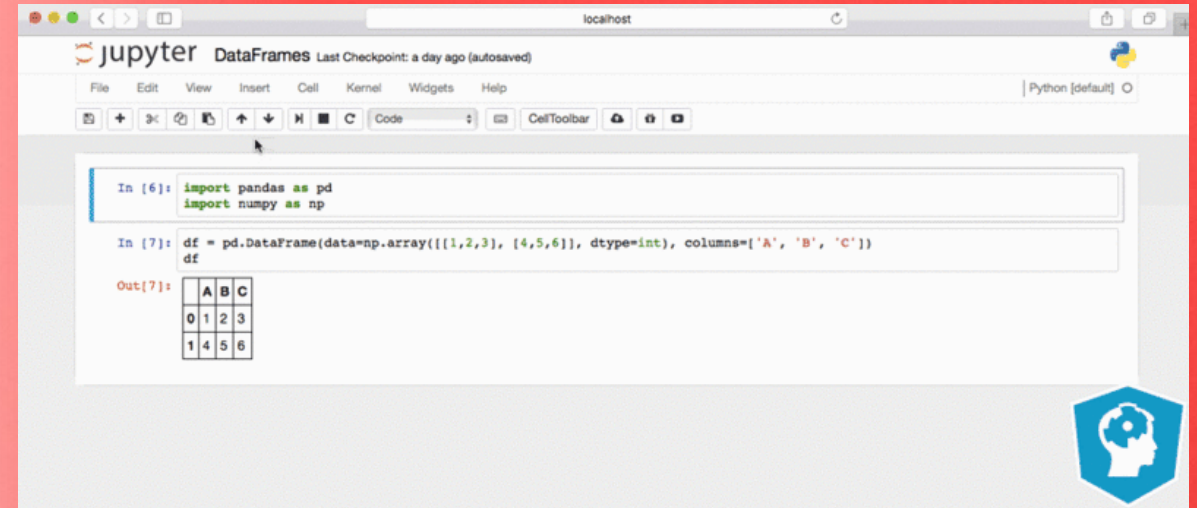


Popular Analytics Tools

Jupyter Notebook is an open-source interactive authoring software.

IT is Open-source and used for sharing code and creating tutorials.

Jupyter Notebook **allows you to create interactive documents and share and work on live code, equations, and visualisations.**

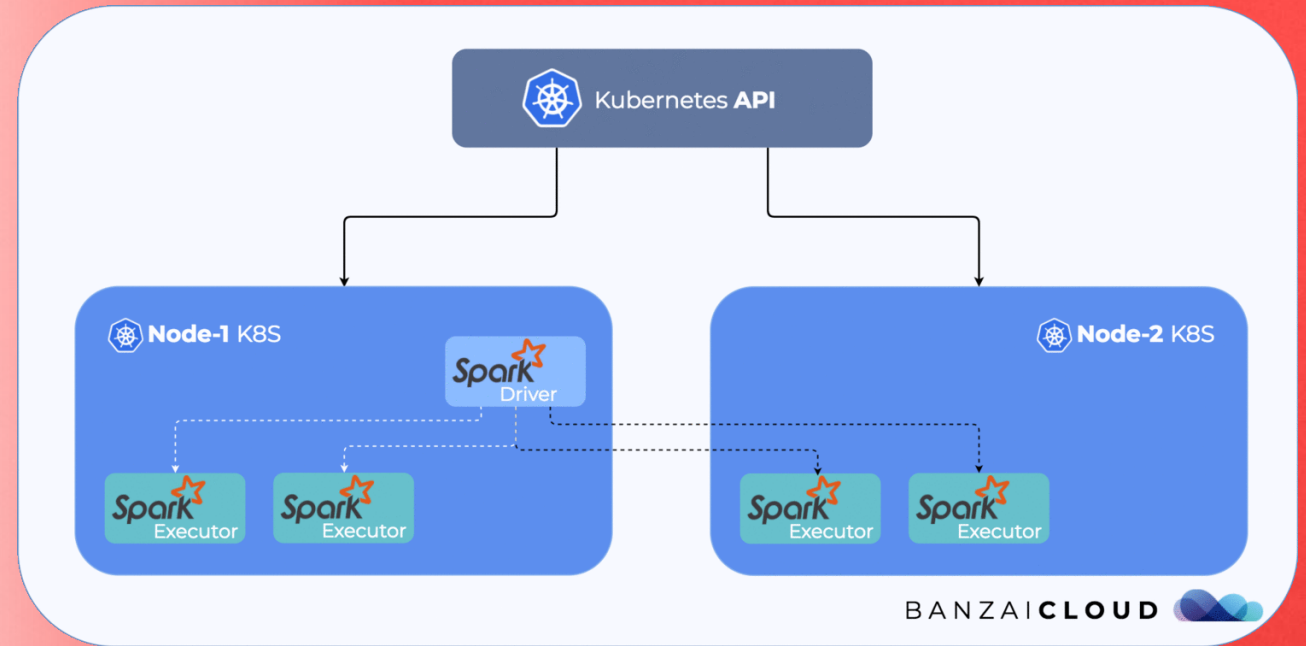


Popular Analytics Tools

Apache Spark is a Data processing framework.

It is **open-source** and mostly used for: Big data processing, analysing **unstructured data**.

It is exceptionally fast because it uses the RAM of computer rather than local memory. This is why it can be up to 100 times faster than similar platforms.

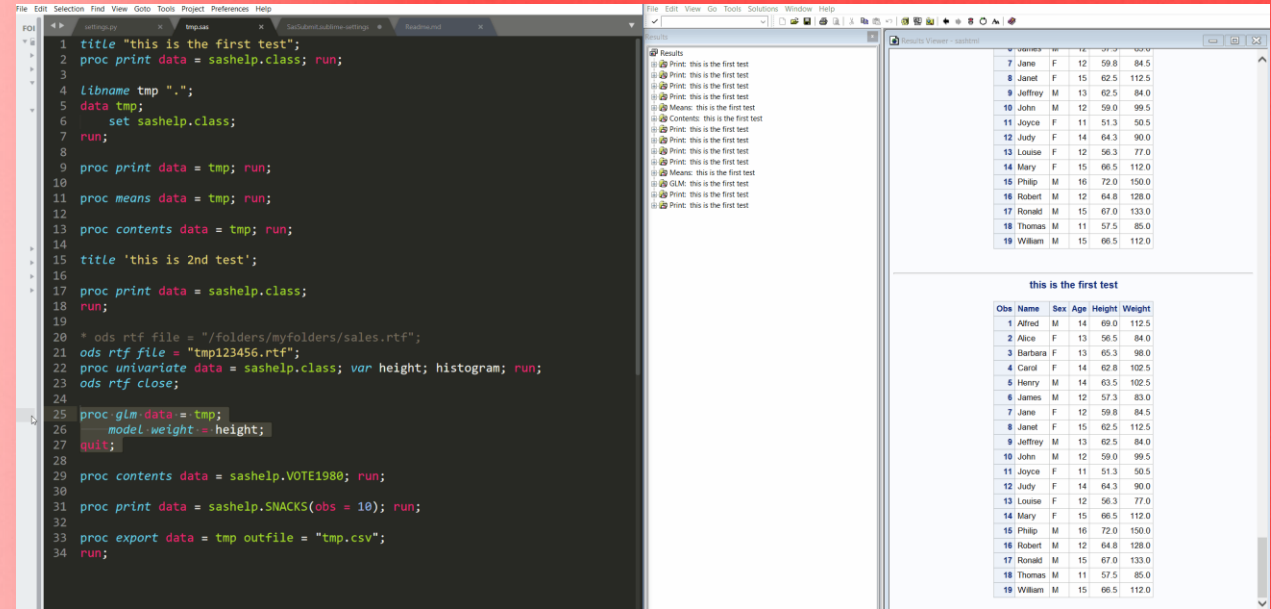


Popular Analytics Tools

It is a statistical software system.

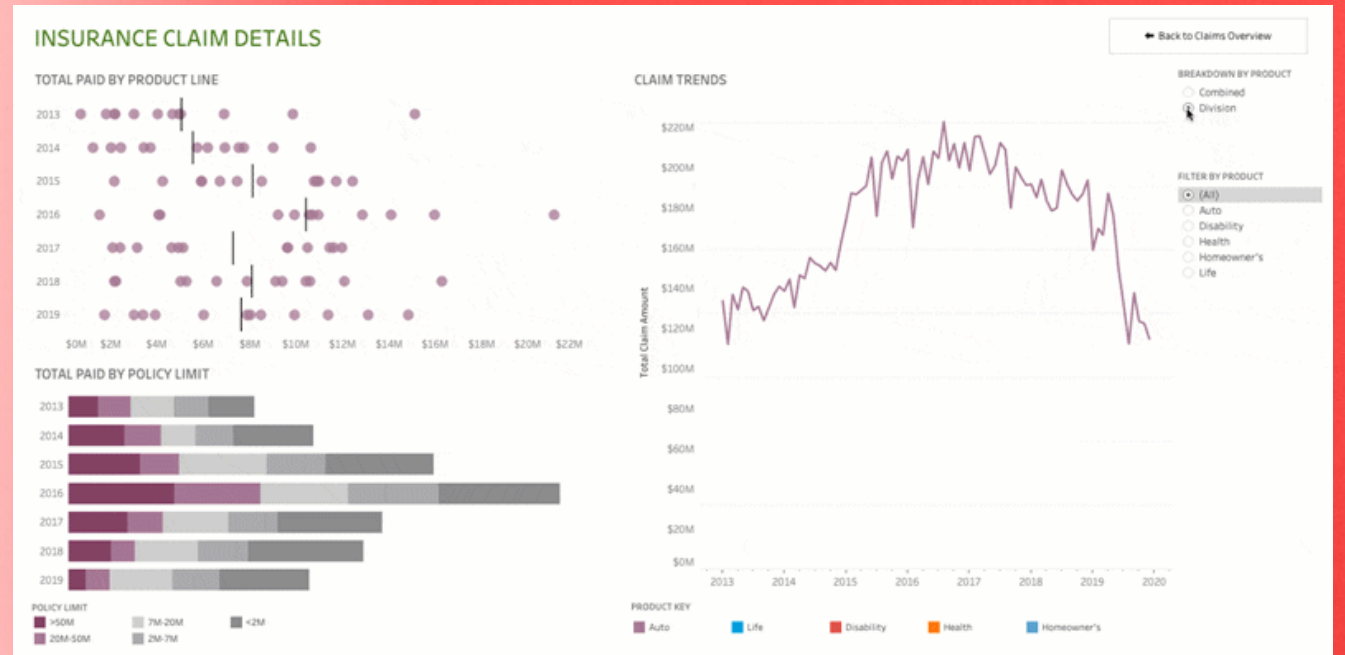
It is not open access, which means
you need to pay for it.

It is mostly used for: **business
intelligence, reporting, data mining,
and predictive modeling.**



Popular Analytics Tools

Similar to Power BI, Tableau is also a data visualisation tool and is great for data dashboarding



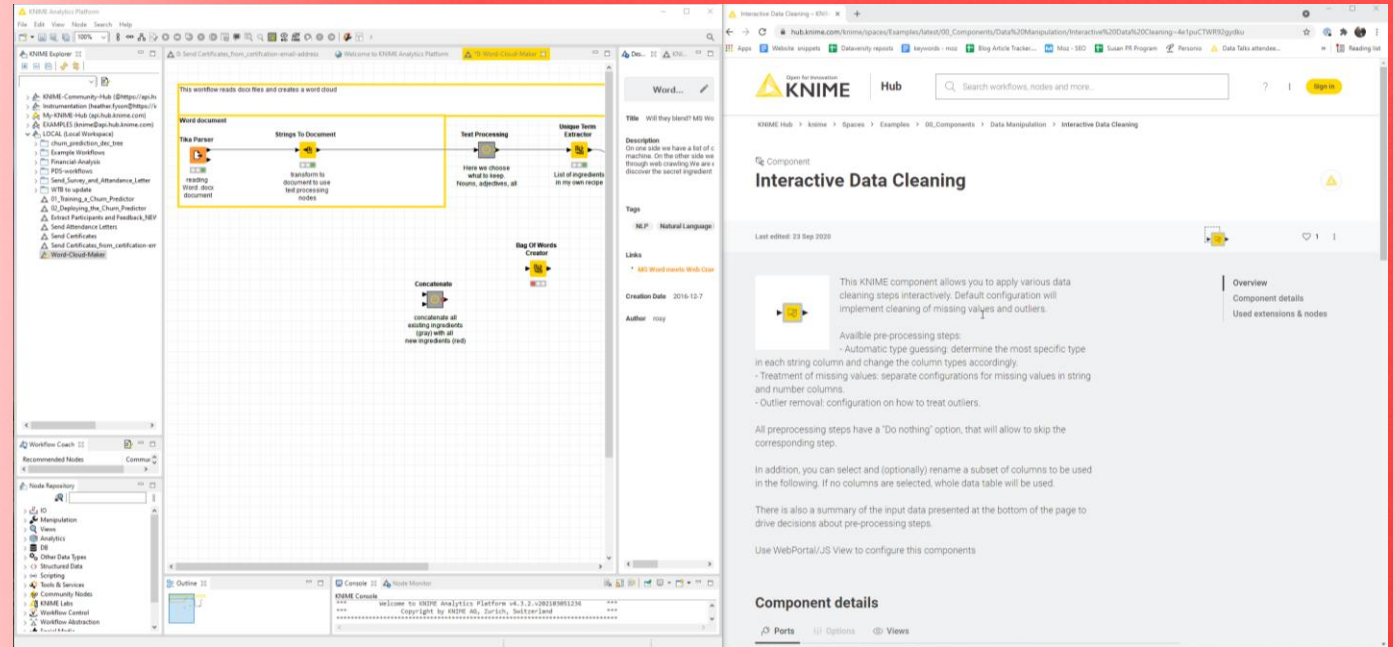
Popular Analytics Tools

Knime is a data integration platform.

It is open-source and is used mostly used for: data mining and machine learning.

Its strength is in importing data from numerous sources into a single one.

And, talking about data importing, this is what we are going to do today using Excel.



What is Supply Chain Analytics?

What is Data Analytics ?



"Systematic analysis of data to obtain meaningful information for better decision making. It involves the collective use of various analytical methodologies"

23

What is a supply chain?

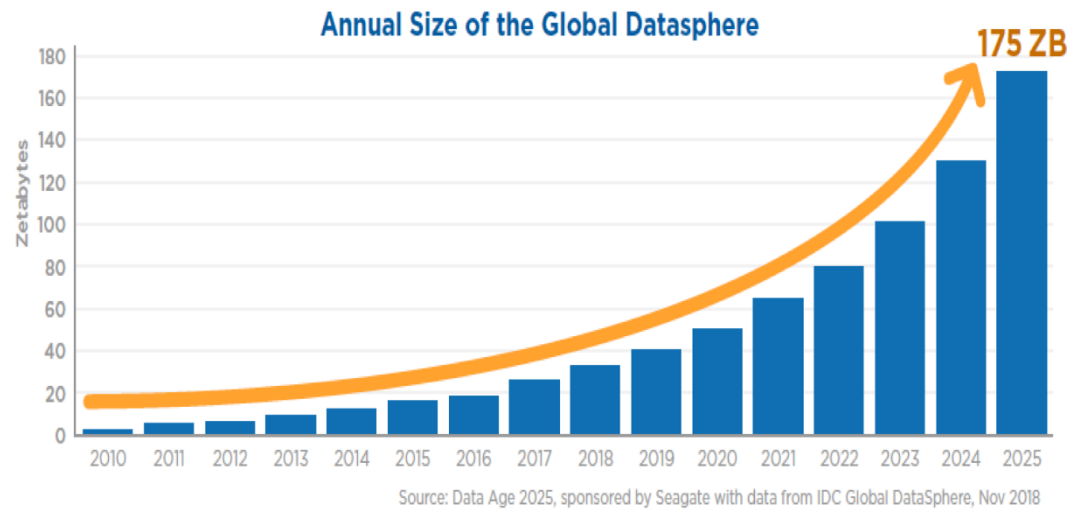
A supply chain refers to the entire process of making and selling products starting from raw materials and components until the products (goods or services) are complete and delivered to the customer.

It includes a network of organisations, people, activities, information, and resources.

Supply chain management is the process of managing activities involved in ordering and receiving raw materials, producing the goods or services, evaluating the quality, packaging, scheduling, delivering, invoicing, and receiving the payments.

Effective supply chain management is about minimising costs and wastages to increase efficiency and profitability of the chain.

22



Abbreviation	Unit	Value	Size (in bytes)
b	bit	0 or 1	1/8 of a byte
B	bytes	8 bits	1 byte
KB	kilobytes	1,000 bytes	1,000 bytes
MB	megabyte	1,000 ² bytes	1,000,000 bytes
GB	gigabyte	1,000 ³ bytes	1,000,000,000 bytes
TB	terabyte	1,000 ⁴ bytes	1,000,000,000,000 bytes
PB	petabyte	1,000 ⁵ bytes	1,000,000,000,000,000 bytes
EB	exabyte	1,000 ⁶ bytes	1,000,000,000,000,000,000 bytes
ZB	zettabyte	1,000 ⁷ bytes	1,000,000,000,000,000,000,000 bytes

Third Party Logistics (3PL)

Third-party logistics (or 3PL) is when some logistics processes are outsourced to a third-party business (including inventory management, warehousing, and fulfilment*).

*Order fulfilment: The complete process from point of sales enquiry to delivery of a product to the customer

Some of the many different services provided by 3PLs are outlined below.

- Transportation
- Warehousing and Inventory management
- Pick and pack – for example picking multiple different SKUs and packing these into single units
- Light manufacturing – acting as contract manufacturers for OEMs, this is quite prevalent in, for example, the electronics sector
- Customs clearance
- Managing reverse logistics.



Third Party Logistics

Downside:

- Less control
- Potential customer loss if improperly managed

Challenges:

- Costs
- Confidentiality
- Performance metrics



FOURTH-PARTY LOGISTICS

Fourth-party logistics (4PL) happens where the entire logistics function of an organisation is outsourced to a logistics service provider (LSP). They often contract with some 3PL.

Unlike third-party logistics (3PL) providers, who typically handle specific logistics functions such as transportation or warehousing, 4PL providers take on a broader role by managing and coordinating the entire logistics of the supply chain on behalf of their clients.

PACK & SEND	
3PL vs 4PL	
3PL	4PL
✓ A 3PL manages only transportation & storage in the supply chain.	✓ A 4PL manages several supply chain vendors for efficiency & cost-effectiveness.
✓ 3PL is suitable for small and medium sized businesses.	✓ 4PL is suitable for medium and large enterprises.
✓ A company may have to contact several 3PL partners for different operations.	✓ The 4PL is the single contact point between a company and different 3PL vendors.
✓ 3PL companies own logistics assets for warehousing and shipping.	✓ A 4PL uses the assets of its 3PL vendors.



How to
import data
from a
CSV/Text file

Let's import data
from this CSV
Text File source to
Excel.

Data cleaning: Delimiter

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

☒ Delimited - Characters such as commas or tabs separate each field.

☐ Fixed width - Fields are aligned in columns with spaces between each field.

Preview of selected data:

1	Column1
2	YearQuarterMonthCityRevenue
3	20171JanuaryPortland 79
4	20171FebruaryPortland148
5	20171MarchPortland 83

Cancel < Back **Next >** Finish

Data cleaning

Now to clean this, first insert a column

B	C	D	E	F	G
	Column1				
	Year	Quarter	Month	City	Revenue
	2017,	1	January	Portland	79
	2017 =	1	February	Portland	148
	2017	1	March	Portland	83
	2017	2	April	Portland	53
	2017	2	May	Portland	61
	2017	2	June	Portland	53
	2017	3	July	Portland	53
	2017	3	August	Portland	66
	2017	3	September	Portland	55
	2017	4	October	Portland	76
	2017	4	November	Portland	50
	2017	4	December	Portland	112
	2018	1	January	Portland	90
	2018	1	February	Portland	39
	2018	1	March	Portland	113
	2018	2	April	Portland	95
	2018	2	May	Portland	51
	2018	2	June	Portland	64
	2018	3	July	Portland	45
	2018	3	August	Portland	90

B	C	D	E	F	G	H	I	J
	Year		Month	City	Revenue.1	Revenue.2	Revenue (Cleaned)	Revenue Values in AUD
	2017		January	Portland	0	0	0	0
	2017		February	Portland	0	0	0	0
	2017		March	Port	20	0	20	27.4
	2017		April	Portland	53	0	53	72.61
	2017		May	Portland	0	61	61	83.57
	2017		June	Portland	53	0	53	72.61
	2017		July	Portland	53	0	53	72.61
	2017		August	Portland	66	0	66	90.42
	2017		September	Portland	55	0	55	75.35
	2017		October	Portland	76	0	76	104.12
	2017		November	Portland	50	0	50	68.5
	2017		December	Portland	112	0	112	153.44
	2017		January	Portland	90	0	90	123.3
	2017		February	Portland	39	0	39	53.43
	2017		March	Portland	113	0	113	154.81
	2017		April	Portland	95	0	95	130.15
	2017		May	Portland	51	0	51	69.87
	2017		June	Portland	64	0	64	87.68
	2017		July	Portland	45	0	45	61.65
	2017		August	Portland	0	90	90	123.3
	2017		September	Portland	116	0	116	158.92

Data cleaning

How to select the entire data, but not the entire sheet?

	A	B	C	D	E
1	Year	Quarter	Month	City	Revenue
2	2017	1	January	Portland	
3	2017	1	February	Portland	
4	2017	1	March	Port	20
5	2017	2	April	Portland	53
6	2017	2	May	Portland	\$61
7	2017	2	June	Portland	53
8	2017	3	July	Portland	53
9	2017	3	August	Portland	66
10	2017	3	September	Portland	55
11	2017	4	October	Portland	76

Ctrl + Shift + ↓

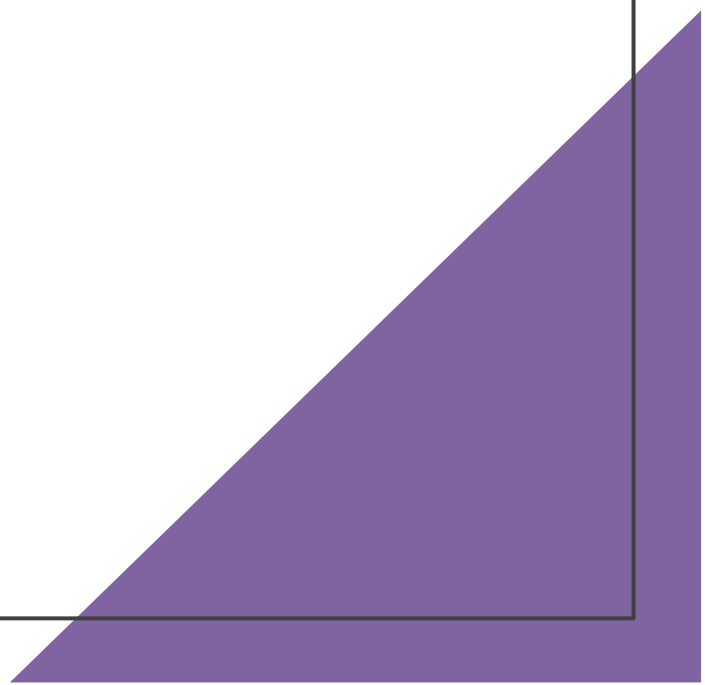
2

Ctrl + Shift + →

3

Removing Duplicated Data

Data Validation

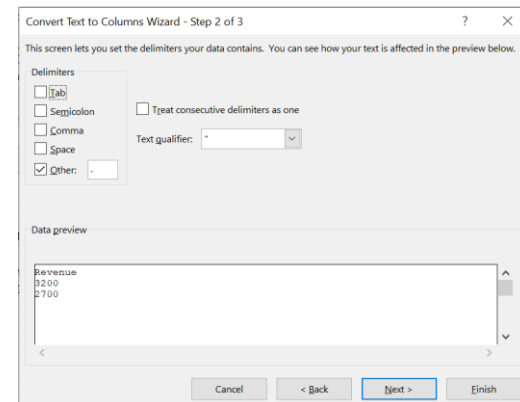


You can
clean find
some
dirtiness at
the top or
bottom of
the column

These can also be cleaned using Data/Text to Column/Delimited Function (you select \$ or AUD for data to be delimited with)

You can also use find and replace function to clean these few examples of data dirtiness

As there are only a few, you can also clean them manually one by one



Missing Value Cells in a Dataset

E	F	G	J	K	L	M
Job Satisf:	Year with	Salary	Working	Performance	Performance Level 2024	Having passion
2	3	100000	35	5	6	€
3	1	90000	38	5	7	5
4	1	110000	42	6	6	€
5	3		52	6	7	€
5	4	115000	56	6	6	7
3	7	140000	60	5		5
4	2	80000	35	6	5	7
3	1	65000	40	6	6	7
2	4	100000	50	6	6	€
1	3	85000	42	7	5	€

If the value of parameters which you are going to investigate are missing, you can put **the average or median of that column** there, but we can also be smarter.

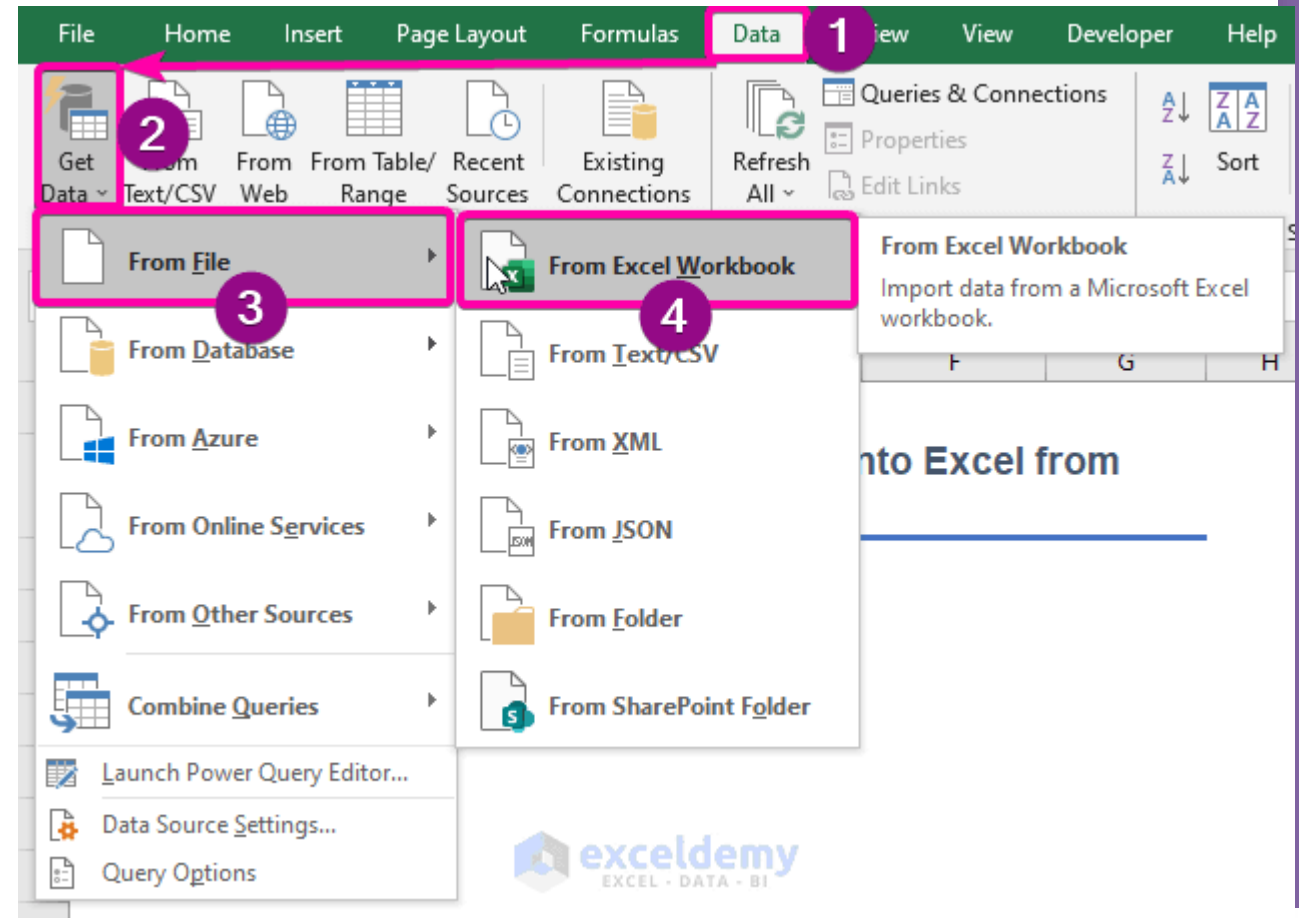
Missing Value Cells in a Dataset

E	F	G	J	K	L	M
Job Satisf	Year with	Salary	Working	Performance	Performance Level 2024	Having passion
2	3	100000	35	5	6	€
3	1	90000	38	5	7	5
4	1	110000	42	6	6	€
5	3	110000	52	6	7	€
5	4	115000	56	6	6	7
3	7	140000	60	5	5	5
4	2	80000	35	6	5	7
3	1	65000	40	6	6	7
2	4	100000	50	6	6	€
1	3	85000	42	7	5	€

What does this mean?

For instance, if you have four production lines, and a **person's performance level** is missing, you should replace this missing value with the average or mean of **people who are working in that production line**, not all people in the factory. This is more likely to be a better representative of the missing value. 😊

Importing data from other sources



Importing Data From Web

